

Sayeef Salahuddin

professor

基本信息

姓名: Sayeef Salahuddin

所属院校: 加州大学-伯克利

地区: United States

年龄范围: 40岁至50岁

海外经历: 有

荣誉称号

★ APS Fellow

★ IEEE Fellow

这些成就代表了在学术领域的最高认可，展现了卓越的研究能力和影响力。

学科领域

微电子科学与工程

材料物理

集成电路设计与集成系统

研究领域

微电子器件与制造

纳米电子与先进工艺

薄膜材料与表面物理

磁性材料与磁学

EDA工具与电路验证

低功耗电路与能量管理

个人简介

Sayeef Salahuddin is the TSMC Distinguished Professor of Electrical Engineering and Computer Sciences at the University of California, Berkeley. He earned his B.S. from Bangladesh University of Engineering and Technology and his Ph.D. from Purdue University. Salahuddin's research focuses on novel device physics for low-power electronics and spintronics, leading to the

discovery of Negative Capacitance Transistors. He has received numerous awards, including the Presidential Early Career Award for Scientists and Engineers, and is a fellow of the IEEE and APS. Salahuddin directs the Berkeley Device Modeling Center and the Berkeley Center for Negative Capacitance Transistors.

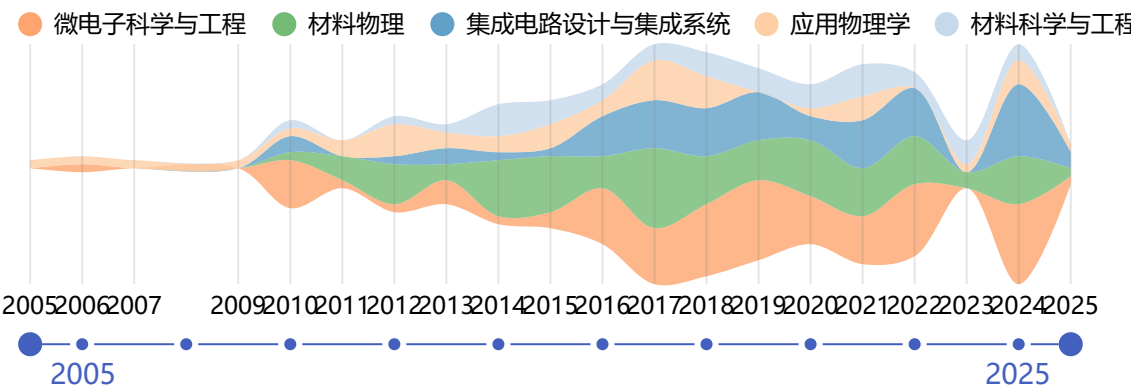
工作经历

University of California, Berkeley - Professor	2008 - 至今
IEEE Electron Devices Letters - Editorial Board Member	2013-01 - 2016-12
IEEE Electron Devices Society committee on Nanotechnology - Chair	2014-01 - 2016-12

教育经历

Purdue University - PhD	- 2007
Bangladesh University of Engineering and Technology - BSc	- 2003

近年发表的论文



Ultra-Thin Body, Short Channel Silicon Transistors Down to 3-nm Si Channel
IEEE ELECTRON DEVICE LETTERS
Park, Jong Ho; Wang, Li-Chen; Sikder, Urmita; Hsu, Shang-Lin; Lee, Chia-Chun; Garg, Chirag; Shanker, Nirmaan; Cheema, Suraj; Li, Wenshen; Hu, Chenming; Salahuddin, Sayeef
被引用次数: 0 2025

Boltzmann-Statistics-Aware Non-Quasi-Static-Charge Model for IC Simulations
IEEE TRANSACTIONS ON ELECTRON DEVICES
Dabhi, Chetan Kumar; Pahwa, Girish; Salahuddin, Sayeef; Hu, Chenming

Anisotropic Ferroelectricity in Polar Vortices

ADVANCED MATERIALS

Behera, Piush; Ross, Aiden M.; Shanker, Nirmaan; Meisenheimer, Peter; Manna, Mahir; Lin, Ching-Che; Hsu, Shang-Lin; Harris, Isaac; Kavle, Pravin; Husain, Sajid; Ojha, Shashank K.; Das, Sujit; Raja, Archana; Martin, Lane W.; Susarla, Sandhya; Salahuddin, Sayeef; Chen, Long-Qing; Ramesh, Ramamoorthy

被引用次数: 0

2025

Topological edge conduction induced by strong anisotropic exchange interactions

PHYSICAL REVIEW B

Sayed, Shehrin; Brahma, Pratik; Hsu, Cheng-Hsiang; Salahuddin, Sayeef

被引用次数: 1

2024

On the coupling between magnetoelastic waves and nitrogen vacancy centers in diamond

APPLIED PHYSICS LETTERS

Jung, Adi; Margueron, Samuel; Bartasyte, Ausrine; Salahuddin, Sayeef

被引用次数: 0

2024

Emergent classical spin liquid phases in an Ising lattice via size effects

PHYSICAL REVIEW B

Brahma, Pratik; Salahuddin, Sayeef

被引用次数: 0

2024

A versatile compact model of resistive random-access memory (RRAM)

SOLID-STATE ELECTRONICS

Tung, Chien-Ting; Dabhi, Chetan Kumar; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 0

2024

Manipulating chiral spin transport with ferroelectric polarization

NATURE MATERIALS

Huang, Xiaoxi; Chen, Xianzhe; Li, Yuhang; Mangeri, John; Zhang, Hongrui; Ramesh, Maya; Taghinejad, Hossein; Meisenheimer, Peter; Caretta, Lucas; Susarla, Sandhya; Jain, Rakshit; Klewe, Christoph; Wang, Tianye; Chen, Rui; Hsu, Cheng-Hsiang; Harris, Isaac; Husain, Sajid; Pan, Hao; Yin, Jia; Shafer, Padraic; Qiu, Ziqiang; Rodrigues, Davi R.; Heinonen, Olle; Vasudevan, Dilip; Iniguez, Jorge; Schlom, Darrell G.; Salahuddin, Sayeef; Martin, Lane W.; Analytis, James G.; Ralph, Daniel C.; Cheng, Ran; Yao, Zhi; Ramesh, Ramamoorthy

被引用次数: 1

2024

Anisotropic Ferroelectricity in Polar Vortices

ADVANCED MATERIALS

Behera, Piush; Ross, Aiden M.; Shanker, Nirmaan; Meisenheimer, Peter; Manna, Mahir; Lin, Ching-Che; Hsu, Shang-Lin; Harris, Isaac; Kavle, Pravin; Husain, Sajid; Ojha, Shashank K.; Das, Sujit; Raja, Archana; Martin, Lane W.; Susarla, Sandhya; Salahuddin, Sayeef; Chen, Long-Qing; Ramesh, Ramamoorthy

被引用次数: 0

2024

Probing the Thermal and Electrical Properties of Ultrawide Bandgap Nitrogen-Polar AlGaN Heterostructures

ADVANCED FUNCTIONAL MATERIALS

Noshin, Maliha; Kwon, Heungdong; Khan, Asir Intisar; Alaei, Sauviz P.; Meng, Chuanzhe; Asheghi, Mehdi; Suzuki, Yuri; Salahuddin, Sayeef; Goodson, Kenneth; Chowdhury, Srabanti

被引用次数: 0

2024

3D Ferroelectric Phase Field Simulations of Polycrystalline Multi-Phase Hafnia and Zirconia Based Ultra-Thin Films

ADVANCED ELECTRONIC MATERIALS

Kumar, Prabhat; Hoffmann, Michael; Nonaka, Andy; Salahuddin, Sayeef; Yao, Zhi (Jackie)

被引用次数: 1

2024

Roadmap on low-power electronics

APL MATERIALS

Ramesh, Ramamoorthy; Salahuddin, Sayeef; Datta, Suman; Diaz, Carlos H.; Nikonov, Dmitri E.; Young, Ian A.; Ham, Donhee; Chang, Meng-Fan; Khwa, Win-San; Lele, Ashwin Sanjay; Binek, Christian; Huang, Yen-Lin; Sun, Yuan-Chen; Chu, Ying-Hao; Prasad, Bhagwati; Hoffmann, Michael; Hu, Jia-Mian; Yao, Zhi (Jackie); Bellaiche, Laurent; Wu, Peng; Cai, Jun; Appenzeller, Joerg; Datta, Supriyo; Camsari, Kerem Y.; Kwon, Jaesuk; Incorvia, Jean Anne C.; Asselberghs, Inge; Ciubotaru, Florin; Couet, Sebastien; Adelman, Christoph; Zheng, Yi; Lindenberg, Aaron M.; Evans, Paul G.; Ercius, Peter; Radu, Iuliana P.

被引用次数: 0

2024

A Novel Neural Network-Based Transistor Compact Model Including Self-Heating

IEEE ELECTRON DEVICE LETTERS

Tung, Chien-Ting; Pampori, Ahtisham; Dabhi, Chetan Kumar; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 0

2024

Non-Quasi-Static Modeling of Neural Network-Based Transistor Compact Model for Fast Transient, AC, and RF Simulations

IEEE ELECTRON DEVICE LETTERS

Tung, Chien-Ting; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 0

2024

Boost in Carrier Velocity Due to Electrostatic Effects of Negative Capacitance Gate Stack

IEEE ELECTRON DEVICE LETTERS

Garg, Chirag; Cheema, Suraj; Shanker, Nirmaan; Li, Wenshen; Hu, Chenming; Salahuddin, Sayeef

被引用次数: 0

2024

An All-Region BSIM Thin-Film Transistor Model for Display and BEOL 3-D Integration Applications

IEEE TRANSACTIONS ON ELECTRON DEVICES

Pahwa, Girish; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 0

2024

Symmetric BSIM-SOI-Part I: A Compact Model for Dynamically Depleted SOI MOSFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Dabhi, Chetan Kumar; Rajasekharan, Dinesh; Pahwa, Girish; Nandi, Debashish; Karumuri, Naveen; Turuvekere, Sreenidhi; Dutta, Anupam; Swaminathan, Balaji; Srihari, Srikanth; Chauhan, Yogesh S.; Salahuddin, Sayeef; Hu, Chenming

Symmetric BSIM-SOI-Part II: A Compact Model for Partially Depleted SOI MOSFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Dabhi, Chetan Kumar; Nandi, Debashish; Nandan, Keshari; Rajasekharan, Dinesh; Pahwa, Girish; Karumuri, Naveen; Turuvekere, Sreenidhi; Dutta, Anupam; Swaminathan, Balaji; Srihari, Srikanth; Chauhan, Yogesh S.; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 1

2024

A Compact Model of Perpendicular Spin-Transfer-Torque Magnetic Tunnel Junction

IEEE TRANSACTIONS ON ELECTRON DEVICES

Tung, Chien-Ting; Dasgupta, Avirup; Agarwal, Harshit; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 1

2024

Giant energy storage and power density negative capacitance superlattices

NATURE

Cheema, Suraj S.; Shanker, Nirmaan; Hsu, Shang-Lin; Schaadt, Joseph; Ellis, Nathan M.; Cook, Matthew; Rastogi, Ravi; Pilawa-Podgurski, Robert C. N.; Ciston, Jim; Mohamed, Mohamed; Salahuddin, Sayeef

被引用次数: 14

2024

Tunable multistate field-free switching and ratchet effect by spin-orbit torque in canted ferrimagnetic alloy

NATURE COMMUNICATIONS

Hsu, Cheng-Hsiang; Gross, Miela J.; Kleidermacher, Hannah Calzi; Sayed, Shehrin; Salahuddin, Sayeef

被引用次数: 0

2024

Non-volatile magnon transport in a single domain multiferroic

NATURE COMMUNICATIONS

Husain, Sajid; Harris, Isaac; Meisenheimer, Peter; Mantri, Sukriti; Li, Xinyan; Ramesh, Maya; Behera, Piush; Taghinejad, Hossein; Kim, Jaegy; Kavle, Pravin; Zhou, Shiyu; Kim, Tae Yeon; Zhang, Hongrui; Stevenson, Paul; Analytis, James G.; Schlom, Darrell; Salahuddin, Sayeef; Iniguez-Gonzalez, Jorge; Xu, Bin; Martin, Lane W.; Caretta, Lucas; Han, Yimo; Bellaiche, Laurent; Yao, Zhi; Ramesh, Ramamoorthy

被引用次数: 2

2024

A SPICE-compatible Neural Network Compact Model for Efficient IC Simulations

2024 INTERNATIONAL CONFERENCE ON SIMULATION OF SEMICONDUCTOR PROCESSES AND DEVICES, SISPAD 2024

Tung, Chien-Ting; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 0

2024

Topological phases in polar oxide nanostructures

REVIEWS OF MODERN PHYSICS

Junquera, Javier; Nahas, Yousra; Prokhorenko, Sergei; Bellaiche, Laurent; iniguez, Jorge; Schlom, Darrell G.; Chen, Long-Qing; Salahuddin, Sayeef; Muller, David A.; Martin, Lane W.; Ramesh, R.

被引用次数: 47

2023

FerroX: A GPU-accelerated, 3D phase-field simulation framework for modeling ferroelectric devices

COMPUTER PHYSICS COMMUNICATIONS

Kumar, Prabhat; Nonaka, Andrew; Jambunathan, Revathi; Pahwa, Girish; Salahuddin, Sayeef; Yao, Zhi (Jackie)

被引用次数: 8

2023

In-Situ Measurement of Magnetoelectric Coupling and Strain Transfer in Multiferroic Nanocomposites of CoFe₂O₄ and Hf_{0.5}Zr_{0.5}O₂ with Residual Porosity

NANO LETTERS

Patel, Shreya K.; Robertson, Daniel D.; Cheema, Suraj S.; Salahuddin, Sayeef; Tolbert, Sarah H.

被引用次数: 4

2023

Roadmap on ferroelectric hafnia- and zirconia-based materials and devices

APL MATERIALS

Silva, Jose P. B.; Alcala, Ruben; Avci, Uygur E.; Barrett, Nick; Begon-Lours, Laura; Borg, Mattias; Byun, Seungyong; Chang, Sou-Chi; Cheong, Sang-Wook; Choe, Duk-Hyun; Coignus, Jean; Deshpande, Veeresh; Dimoulas, Athanasios; Dubourdieu, Catherine; Fina, Ignasi; Funakubo, Hiroshi; Grenouillet, Laurent; Gruverman, Alexei; Heo, Jinseong; Hoffmann, Michael; Hsain, H. Alex; Huang, Fei-Ting; Hwang, Cheol Seong; Iniguez, Jorge; Jones, Jacob L.; Karpov, Ilya V.; Kersch, Alfred; Kwon, Taegyu; Lancaster, Suzanne; Lederer, Maximilian; Lee, Younghwan; Lomenzo, Patrick D.; Martin, Lane W.; Martin, Simon; Migita, Shinji; Mikolajick, Thomas; Noheda, Beatriz; Park, Min Hyuk; Rabe, Karin M.; Salahuddin, Sayeef; Sanchez, Florencio; Seidel, Konrad; Shimizu, Takao; Shiraishi, Takahisa; Slesazeck, Stefan; Toriumi, Akira; Uchida, Hiroshi; Vilquin, Bertrand; Xu, Xianghan; Ye, Kun Hee; Schroeder, Uwe

被引用次数: 43

2023

Accelerated Ultrafast Magnetization Dynamics at Graphene/CoGd Interfaces

ACS NANO

Mondal, Sucheta; Lin, Yuxuan; Polley, Debanjan; Su, Cong; Zettl, Alex; Salahuddin, Sayeef; Bokor, Jeffrey

被引用次数: 5

2022

RKKY Exchange Bias Mediated Ultrafast All-Optical Switching of a Ferromagnet

ADVANCED FUNCTIONAL MATERIALS

Chatterjee, Jyotirmoy; Polley, Debanjan; Pattabi, Akshay; Jang, Hyejin; Salahuddin, Sayeef; Bokor, Jeffrey

被引用次数: 20

2022

One Nanometer HfO₂-Based Ferroelectric Tunnel Junctions on Silicon

ADVANCED ELECTRONIC MATERIALS

Cheema, Suraj S.; Shanker, Nirmaan; Hsu, Cheng-Hsiang; Datar, Adhiraj; Bae, Jongho; Kwon, Daewoong; Salahuddin, Sayeef

被引用次数: 80

2022

Logically synthesized and hardware-accelerated restricted Boltzmann machines for combinatorial optimization and integer factorization

NATURE ELECTRONICS

Patel, Saavan; Canoz, Philip; Salahuddin, Sayeef

被引用次数: 24

2022

Write Disturb-Free Ferroelectric FETs With Non-Accumulative Switching Dynamics

IEEE ELECTRON DEVICE LETTERS

Hoffmann, Michael; Tan, Ava Jiang; Shanker, Nirmaan; Liao, Yu-Hung; Wang, Li-Chen; Bae, Jong-Ho; Hu, Chenming; Salahuddin, Sayeef

被引用次数: 9

2022

A Compact Model of Ferroelectric Field-Effect Transistor

IEEE ELECTRON DEVICE LETTERS

Tung, Chien-Ting; Pahwa, Girish; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 11

2022

Fast Read-After-Write and Depolarization Fields in High Endurance n-Type Ferroelectric FETs

IEEE ELECTRON DEVICE LETTERS

Hoffmann, Michael; Tan, Ava Jiang; Shanker, Nirmaan; Liao, Yu-Hung; Wang, Li-Chen; Bae, Jong-Ho; Hu, Chenming; Salahuddin, Sayeef

被引用次数: 31

2022

Large Injection Velocities in Highly Scaled, Fully Depleted Silicon on Insulator Transistors

IEEE ELECTRON DEVICE LETTERS

Liao, Yu-Hung; Aabrar, Khandker Akif; Chakraborty, Wriddhi; Li, Wenshen; Datta, Suman; Salahuddin, Sayeef

被引用次数: 4

2022

A Compact Model of Antiferroelectric Capacitor

IEEE ELECTRON DEVICE LETTERS

Tung, Chien-Ting; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 4

2022

Dynamical Effects of Excess Carriers on SOI FeFET Memory Device Operations

IEEE TRANSACTIONS ON ELECTRON DEVICES

Hsia, Jung-Han; Liao, Yu-Hung; Ramachandran, Neelesh; Salahuddin, Sayeef

被引用次数: 0

2022

Critical Importance of Nonuniform Polarization and Fringe Field Effects for Scaled Ferroelectric FinFET Memory

IEEE TRANSACTIONS ON ELECTRON DEVICES

Pahwa, Girish; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 4

2022

A Compact Model of Nanoscale Ferroelectric Capacitor

IEEE TRANSACTIONS ON ELECTRON DEVICES

Tung, Chien-Ting; Pahwa, Girish; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 3

2022

A Compact Model of Metal-Ferroelectric-Insulator-Semiconductor Tunnel Junction

IEEE TRANSACTIONS ON ELECTRON DEVICES

Tung, Chien-Ting; Pahwa, Girish; Salahuddin, Sayeef; Hu, Chenming

Emergent ferroelectricity in subnanometer binary oxide films on silicon

SCIENCE

Cheema, Suraj S.; Shanker, Nirmaan; Hsu, Shang-Lin; Rho, Yoonsoo; Hsu, Cheng-Hsiang; Stoica, Vladimir A.; Zhang, Zhan; Freeland, John W.; Shafer, Padraic; Grigoropoulos, Costas P.; Ciston, Jim; Salahuddin, Sayeef

被引用次数: 102

2022

Ultrathin ferroic HfO₂-ZrO₂ superlattice gate stack for advanced transistors

NATURE

Cheema, Suraj S.; Shanker, Nirmaan; Wang, Li-Chen; Hsu, Cheng-Hsiang; Hsu, Shang-Lin; Liao, Yu-Hung; San Jose, Matthew; Gomez, Jorge; Chakraborty, Wriddhi; Li, Wenshen; Bae, Jong-Ho; Volkman, Steve K.; Kwon, Daewoong; Rho, Yoonsoo; Pinelli, Gianni; Rastogi, Ravi; Pipitone, Dominick; Stull, Corey; Cook, Matthew; Tyrrell, Brian; Stoica, Vladimir A.; Zhang, Zhan; Freeland, John W.; Tassone, Christopher J.; Mehta, Apurva; Saheli, Ghazal; Thompson, David; Suh, Dong Ik; Koo, Won-Tae; Nam, Kab-Jin; Jung, Dong Jin; Song, Woo-Bin; Lin, Chung-Hsun; Nam, Seunggeol; Heo, Jinseong; Parihar, Narendra; Grigoropoulos, Costas P.; Shafer, Padraic; Fay, Patrick; Ramesh, Ramamoorthy; Mahapatra, Souvik; Ciston, Jim; Datta, Suman; Mohamed, Mohamed; Hu, Chenming; Salahuddin, Sayeef

被引用次数: 157

2022

Unified Framework for Charge-Spin Interconversion in Spin-Orbit Materials

PHYSICAL REVIEW APPLIED

Sayed, Shehrin; Hong, Seokmin; Huang, Xiaoxi; Caretta, Lucas; Everhardt, Arnoud S.; Ramesh, Ramamoorthy; Salahuddin, Sayeef; Datta, Supriyo

被引用次数: 8

2021

Double-peaked resonance in harmonic-free acoustically driven ferromagnetic resonance

APPLIED PHYSICS LETTERS

Jung, Adi; Macri, Dorotea; Margueron, Samuel; Bartasyte, Ausrine; Salahuddin, Sayeef

被引用次数: 3

2021

Spin-orbit torque rectifier for weak RF energy harvesting

APPLIED PHYSICS LETTERS

Sayed, Shehrin; Salahuddin, Sayeef; Yablonovitch, Eli

被引用次数: 5

2021

Unifying femtosecond and picosecond single-pulse magnetic switching in Gd-Fe-Co

PHYSICAL REVIEW B

Jakobs, F.; Ostler, T. A.; Lambert, C-H; Yang, Y.; Salahuddin, S.; Wilson, R. B.; Gorchon, J.; Bokor, J.; Atxitia, U.

被引用次数: 30

2021

Negative capacitance enables GAA scaling VDD to 0.5 V

SOLID-STATE ELECTRONICS

Kao, Ming-Yen; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 10

2021

Local negative permittivity and topological phase transition in polar skyrmions (vol 20, pg 194, 2021)

NATURE MATERIALS

Das, S.; Hong, Z.; Stoica, V. A.; Goncalves, M. A. P.; Shao, Y. T.; Parsonnet, E.; Marksz, E. J.; Saremi, S.; McCarter, M. R.; Reynoso, A.; Long, C. J.; Hagerstrom, A. M.; Meyers, D.; Ravi, V.; Prasad, B.; Zhou, H.; Zhang, Z.; Wen, H.; Gomez-Ortiz, F.; Garcia-Fernandez, P.; Bokor, J.; iniguez, J.; Freeland, J. W.; Orloff, N. D.; Junquera, J.; Chen, L. Q.; Salahuddin, S.; Muller, D. A.; Martin, L. W.; Ramesh, R.

被引用次数: 1

2021

Local negative permittivity and topological phase transition in polar skyrmions

NATURE MATERIALS

Das, S.; Hong, Z.; Stoica, V. A.; Goncalves, M. A. P.; Shao, Y. T.; Parsonnet, E.; Marksz, E. J.; Saremi, S.; McCarter, M. R.; Reynoso, A.; Long, C. J.; Hagerstrom, A. M.; Meyers, D.; Ravi, V.; Prasad, B.; Zhou, H.; Zhang, Z.; Wen, H.; Gomez-Ortiz, F.; Garcia-Fernandez, P.; Bokor, J.; iniguez, J.; Freeland, J. W.; Orloff, N. D.; Junquera, J.; Chen, L. Q.; Salahuddin, S.; Muller, D. A.; Martin, L. W.; Ramesh, R.

被引用次数: 126

2021

Novel Spin-Orbit Torque Generation at Room Temperature in an All-Oxide Epitaxial La_{0.7}Sr_{0.3}MnO₃/SrIrO₃ System

ADVANCED MATERIALS

Huang, Xiaoxi; Sayed, Shehrin; Mittelstaedt, Joseph; Susarla, Sandhya; Karimeddiny, Saba; Caretta, Lucas; Zhang, Hongrui; Stoica, Vladimir A.; Gosavi, Tanay; Mahfouzi, Farzad; Sun, Qilong; Ercius, Peter; Kioussis, Nicholas; Salahuddin, Sayeef; Ralph, Daniel C.; Ramesh, Ramamoorthy

被引用次数: 44

2021

Epitaxial Ferroelectric Hf_{0.5}Zr_{0.5}O₂ with Metallic Pyrochlore Oxide Electrodes

ADVANCED MATERIALS

Zhang, Zimeng; Hsu, Shang-Lin; Stoica, Vladimir A.; Paik, Hanjong; Parsonnet, Eric; Qualls, Alexander; Wang, Jianjun; Xie, Liang; Kumari, Mukesh; Das, Sujit; Leng, Zhinan; McBriarty, Martin; Proksch, Roger; Gruverman, Alexei; Schlom, Darrell G.; Chen, Long-Qing; Salahuddin, Sayeef; Martin, Lane W.; Ramesh, Ramamoorthy

被引用次数: 25

2021

Ferroelectric gate oxides for negative capacitance transistors

MRS BULLETIN

Hoffmann, Michael; Salahuddin, Sayeef

被引用次数: 12

2021

A Voltage-Controlled Gain Cell Magnetic Memory

IEEE ELECTRON DEVICE LETTERS

Sayed, Shehrin; Hsu, Cheng-Hsiang; Salahuddin, Sayeef

被引用次数: 0

2021

Ferroelectric HfO₂ Memory Transistors With High- κ Interfacial Layer and Write Endurance Exceeding 10¹⁰ Cycles

IEEE ELECTRON DEVICE LETTERS

Tan, Ava Jiang; Liao, Yu-Hung; Wang, Li-Chen; Shanker, Nirmaan; Bae, Jong-Ho; Hu, Chenming; Salahuddin, Sayeef

被引用次数: 147

2021

A Compact Model of Polycrystalline Ferroelectric Capacitor

IEEE TRANSACTIONS ON ELECTRON DEVICES

Tung, Chien-Ting; Pahwa, Girish; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 19

2021

Compact Modeling of Temperature Effects in FDSOI and FinFET Devices Down to Cryogenic Temperatures

IEEE TRANSACTIONS ON ELECTRON DEVICES

Pahwa, Girish; Kushwaha, Pragya; Dasgupta, Avirup; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 52

2021

Energy Storage and Reuse in Negative Capacitance

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kao, Ming-Yen; Liao, Yu-Hung; Pahwa, Girish; Dasgupta, Avirup; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 1

2021

Electric Field-Induced Permittivity Enhancement in Negative-Capacitance FET

IEEE TRANSACTIONS ON ELECTRON DEVICES

Liao, Yu-Hung; Kwon, Daewoong; Cheema, Suraj; Shanker, Nirmaan; Tan, Ava J.; Kao, Ming-Yen; Wang, Li-Chen; Hu, Chenming; Salahuddin, Sayeef

被引用次数: 10

2021

Towards the Integration of Hf_{0.8}Zr_{0.2}O₂-based Negative Capacitance Dielectrics on β -Ga₂O₃ Substrates

PROCEEDINGS OF THE 2021 IEEE NATIONAL AEROSPACE AND ELECTRONICS CONFERENCE (NAECON)

Salcedo, Guillermo A.; Islam, Ahmad E.; Dietz, Michael K.; Cheema, Suraj; Leedy, Kevin D.; Liddy, Kyle J.; Green, Andrew J.; Wang, Weisong; Salahuddin, Sayeef; Chabak, Kelson D.; Sattler, James M.

被引用次数: 1

2021

Electric-field control of spin dynamics during magnetic phase transitions

SCIENCE ADVANCES

Nan, Tianxiang; Lee, Yeonbae; Zhuang, Shihao; Hu, Zhongqiang; Clarkson, James D.; Wang, Xinjun; Ko, Changhyun; Choe, HwanSung; Chen, Zuhuang; Budil, David; Wu, Junqiao; Salahuddin, Sayeef; Hu, Jiamian; Ramesh, Ramamoorthy; Sun, Nian

被引用次数: 25

2020

Resonant Enhancement of Exchange Coupling for Voltage-Controlled Magnetic Switching

PHYSICAL REVIEW APPLIED

Sayed, Shehrin; Hsu, Cheng-Hsiang; Roschewsky, Niklas; Yang, See-Hun; Salahuddin, Sayeef

被引用次数: 3

2020

Tunneling electroresistance effects in epitaxial complex oxides on silicon

APPLIED PHYSICS LETTERS

Abuwasib, Mohammad; Serrao, Claudy R.; Stan, Liliana; Salahuddin, Sayeef; Bakaul, Saidur Rahman

被引用次数: 8

2020

Statistically meaningful measure of domain-wall roughness in magnetic thin films

PHYSICAL REVIEW B

Jordan, D.; Albornoz, L. J.; Gorchon, J.; Lambert, C. H.; Salahuddin, S.; Bokor, J.; Curiale, J.; Bustingorry, S.

被引用次数: 10

2020

Ferroelectric Domain Wall Motion in Freestanding Single-Crystal Complex Oxide Thin Film

ADVANCED MATERIALS

Bakaul, Saidur R.; Kim, Jaegyu; Hong, Seungbum; Cherukara, Mathew J.; Zhou, Tao; Stan, Liliana; Serrao, Claudy R.; Salahuddin, Sayeef; Petford-Long, Amanda K.; Fong, Dillon D.; Holt, Martin, V

被引用次数: 37

2020

Tunable Magnetoelastic Effects in Voltage-Controlled Exchange-Coupled Composite Multiferroic Microstructures

ACS APPLIED MATERIALS & INTERFACES

Xiao, Z.; Lo Conte, R.; Goiriena-Goikoetxea, M.; Chopdekar, R. V.; Lambert, C. -H. A.; Li, X.; N'Diaye, A. T.; Shafer, P.; Tiwari, S.; Barra, A.; Chavez, A.; Mohanchandra, K. P.; Carman, G. P.; Wang, K. L.; Salahuddin, S.; Arenholz, E.; Bokor, J.; Candler, R. N.

被引用次数: 19

2020

Fully transparent field-effect transistor with high drain current and on-off ratio

APL MATERIALS

Park, Jisung; Paik, Hanjong; Nomoto, Kazuki; Lee, Kiyoun; Park, Bo-Eun; Grisafe, Benjamin; Wang, Li-Chen; Salahuddin, Sayeef; Datta, Suman; Kim, Yongsung; Jena, Debdeep; Xing, Huili Grace; Schlom, Darrell G.

被引用次数: 26

2020

A Density Metric for Semiconductor Technology

PROCEEDINGS OF THE IEEE

Wong, H. -S. Philip; Akarvardar, Kerem; Antoniadis, Dimitri; Bokor, Jeffrey; Hu, Chenming; King-Liu, Tsu-Jae; Mitra, Subhasish; Plummer, James D.; Salahuddin, Sayeef

被引用次数: 29

2020

Highly Scaled, High Endurance, Ω -Gate, Nanowire Ferroelectric FET Memory Transistors

IEEE ELECTRON DEVICE LETTERS

Bae, Jong-Ho; Kwon, Daewoong; Jeon, Namho; Cheema, Suraj; Tan, Ava Jiang; Hu, Chenming; Salahuddin, Sayeef

被引用次数: 49

2020

Experimental Demonstration of a Ferroelectric HfO₂-Based Content Addressable Memory Cell

IEEE ELECTRON DEVICE LETTERS

Tan, Ava J.; Chatterjee, Korok; Zhou, Jiuren; Kwon, Daewoong; Liao, Yu-Hung; Cheema, Suraj; Hu, Chenming; Salahuddin, Sayeef

被引用次数: 58

2020

Near Threshold Capacitance Matching in a Negative Capacitance FET With 1 nm Effective Oxide Thickness Gate Stack

IEEE ELECTRON DEVICE LETTERS

Kwon, Daewoong; Cheema, Suraj; Lin, Yen-Kai; Liao, Yu-Hung; Chatterjee, Korok; Tan, Ava J.; Hu, Chenming; Salahuddin, Sayeef

被引用次数: 34

2020

Design Optimization Techniques in Nanosheet Transistor for RF Applications

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kushwaha, Pragya; Dasgupta, Avirup; Kao, Ming-Yen; Agarwal, Harshit; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 26

2020

Analysis and Modeling of Polarization Gradient Effect on Negative Capacitance FET

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kao, Ming-Yen; Pahwa, Girish; Dasgupta, Avirup; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 11

2020

BSIM Compact Model of Quantum Confinement in Advanced Nanosheet FETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Dasgupta, Avirup; Parihar, Shivendra Singh; Kushwaha, Pragya; Agarwal, Harshit; Kao, Ming-Yen; Salahuddin, Sayeef; Chauhan, Yogesh Singh; Hu, Chenming

被引用次数: 38

2020

Enhanced ferroelectricity in ultrathin films grown directly on silicon (Expression of Concern of Vol 532, Pg 210, 2016)

NATURE

Cheema, Suraj S.; Kwon, Daewoong; Shanker, Nirmaan; dos Reis, Roberto; Hsu, Shang-Lin; Xiao, Jun; Zhang, Haigang; Wagner, Ryan; Datar, Adhiraj; McCarter, Margaret R.; Serrao, Claudy R.; Yadav, Ajay K.; Karbasian, Golnaz; Hsu, Cheng-Hsiang; Tan, Ava J.; Wang, Li-Chen; Thakare, Vishal; Zhang, Xiang; Mehta, Apurva; Karapetrova, Evguenia; Chopdekar, Rajesh, V; Shafer, Padraic; Arenholz, Elke; Hu, Chenming; Proksch, Roger; Ramesh, Ramamoorthy; Ciston, Jim; Salahuddin, Sayeef

被引用次数: 4

2020

Enhanced ferroelectricity in ultrathin films grown directly on silicon

NATURE

Cheema, Suraj S.; Kwon, Daewoong; Shanker, Nirmaan; dos Reis, Roberto; Hsu, Shang-Lin; Xiao, Jun; Zhang, Haigang; Wagner, Ryan; Datar, Adhiraj; McCarter, Margaret R.; Serrao, Claudy R.; Yadav, Ajay K.; Karbasian, Golnaz; Hsu, Cheng-Hsiang; Tan, Ava J.; Wang, Li-Chen; Thakare, Vishal; Zhang, Xiang; Mehta, Apurva; Karapetrova, Evguenia; Chopdekar, Rajesh, V; Shafer, Padraic; Arenholz, Elke; Hu, Chenming; Proksch, Roger; Ramesh, Ramamoorthy; Ciston, Jim; Salahuddin, Sayeef

被引用次数: 595

2020

A Spin-Orbit-Torque Memristive Device

ADVANCED ELECTRONIC MATERIALS

Zhang, Shuai; Luo, Shijiang; Xu, Nuo; Zou, Qiming; Song, Min; Yun, Jijun; Luo, Qiang; Guo, Zhe; Li, Ruofan; Tian, Weicheng; Li, Xin; Zhou, Hengan; Chen, Huiming; Zhang, Yue; Yang, Xiaofei; Jiang, Wanjun; Shen, Ka; Hong, Jeongmin; Yuan, Zhe; Xi, Li; Xia, Ke; Salahuddin, Sayeef; Dieny, Bernard; You, Long

Micromagnetic analysis and optimization of spin-orbit torque switching processes in synthetic antiferromagnets

JOURNAL OF APPLIED PHYSICS

Tremisina, E. A.; Roschewsky, N.; Salahuddin, S.

被引用次数: 5

2019

Generation and stability of structurally imprinted target skyrmions in magnetic multilayers

APPLIED PHYSICS LETTERS

Kent, Noah; Streubel, Robert; Lambert, Charles-Henri; Ceballos, Alejandro; Je, Soong-Gun; Dhuey, Scott; Im, Mi-Young; Buttner, Felix; Hellman, Frances; Salahuddin, Sayeef; Fischer, Peter

被引用次数: 19

2019

Ultrafast magnetization switching in nanoscale magnetic dots

APPLIED PHYSICS LETTERS

El-Ghazaly, Amal; Tran, Brandon; Ceballos, Alejandro; Lambert, Charles-Henri; Pattabi, Akshay; Salahuddin, Sayeef; Hellman, Frances; Bokor, Jeffrey

被引用次数: 46

2019

Spin-orbit torque and Nernst effect in Bi-Sb/Co heterostructures

PHYSICAL REVIEW B

Roschewsky, Niklas; Walker, Emily S.; Gowtham, Praveen; Muschinske, Sarah; Hellman, Frances; Bank, Seth R.; Salahuddin, Sayeef

被引用次数: 71

2019

Negative Capacitance Transistors

PROCEEDINGS OF THE IEEE

Wong, Justin C.; Salahuddin, Sayeef

被引用次数: 100

2019

Anomalously Beneficial Gate-Length Scaling Trend of Negative Capacitance Transistors

IEEE ELECTRON DEVICE LETTERS

Liao, Yu-Hung; Kwon, Daewoong; Lin, Yen-Kai; Tan, Ava Jiang; Hu, Chenming; Salahuddin, Sayeef

被引用次数: 22

2019

Challenges to Partial Switching of Hf_{0.8}Zr_{0.2}O₂ Gated Ferroelectric FET for Multilevel/Analog or Low-voltage Memory Operation

IEEE ELECTRON DEVICE LETTERS

Chatterjee, Korok; Kim, Sangwan; Karbasian, Golnaz; Kwon, Daewoong; Tan, Ava J.; Yadav, Ajay K.; Serrao, Claudy R.; Hu, Chenming; Salahuddin, Sayeef

被引用次数: 30

2019

Characterization and Modeling of Flicker Noise in FinFETs at Advanced Technology Node

IEEE ELECTRON DEVICE LETTERS

Kushwaha, Pragya; Agarwal, Harshit; Lin, Yen-Kai; Dasgupta, Avirup; Kao, Ming-Yen; Lu, Ye; Yue, Yun; Chen, Xiaonan; Wang, Joseph; Sy, Wing; Yang, Frank; Chidambaram, P. R. Chidi; Salahuddin, Sayeef; Hu,

Chenming

被引用次数: 33

2019

Negative Capacitance FET With 1.8-nm-Thick Zr-Doped HfO₂ Oxide

IEEE ELECTRON DEVICE LETTERS

Kwon, Daewoong; Cheema, Suraj; Shanker, Nirmaan; Chatterjee, Korok; Liao, Yu-Hung; Tan, Ava J.; Hu, Chenming; Salahuddin, Sayeef

被引用次数: 114

2019

Spacer Engineering in Negative Capacitance FinFETs

IEEE ELECTRON DEVICE LETTERS

Lin, Yen-Kai; Agarwal, Harshit; Kao, Ming-Yen; Zhou, Jiuren; Liao, Yu-Hung; Dasgupta, Avirup; Kushwaha, Pragya; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 35

2019

Optimization of NCFET by Matching Dielectric and Ferroelectric Nonuniformly Along the Channel

IEEE ELECTRON DEVICE LETTERS

Kao, Ming-Yen; Lin, Yen-Kai; Agarwal, Harshit; Liao, Yu-Hung; Kushwaha, Pragya; Dasgupta, Avirup; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 22

2019

Proposal for Capacitance Matching in Negative Capacitance Field-Effect Transistors

IEEE ELECTRON DEVICE LETTERS

Agarwal, Harshit; Kushwaha, Pragya; Lin, Yen-Kai; Kao, Ming-Yen; Liao, Yu-Hung; Dasgupta, Avirup; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 69

2019

BSIM-HV: High-Voltage MOSFET Model Including Quasi-Saturation and Self-Heating Effect

IEEE TRANSACTIONS ON ELECTRON DEVICES

Agarwal, H.; Gupta, C.; Goel, R.; Kushwaha, P.; Lin, Y. -K.; Kao, M. -Y.; Duarte, J. -P.; Chang, H. -L.; Chauhan, Y. S.; Salahuddin, S.; Hu, C.

被引用次数: 24

2019

Analysis and Modeling of Inner Fringing Field Effect on Negative Capacitance FinFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lin, Yen-Kai; Agarwal, Harshit; Kushwaha, Pragya; Kao, Ming-Yen; Liao, Yu-Hung; Chatterjee, Korok; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 38

2019

Spatially resolved steady-state negative capacitance

NATURE

Yadav, Ajay K.; Nguyen, Kayla X.; Hong, Zijian; Garcia-Fernandez, Pablo; Aguado-Puente, Pablo; Nelson, Christopher T.; Das, Sujit; Prasad, Bhagawati; Kwon, Daewoong; Cheema, Suraj; Khan, Asif I.; Hu, Chenming; Iniguez, Jorge; Junquera, Javier; Chen, Long-Qing; Muller, David A.; Ramesh, Ramamoorthy; Salahuddin, Sayeef

Novel Cascadable Magnetic Majority Gates for Implementing Comprehensive Logic Functions

IEEE TRANSACTIONS ON ELECTRON DEVICES

Li, Xin; Song, Min; Xu, Nuo; Luo, Shijiang; Zou, Qiming; Zhang, Shuai; Hong, Jeongmin; Yang, Xiaofei; Min, Tai; Han, Xiufeng; Zou, Xuecheng; Zhu, Jian-Gang; Salahuddin, Sayeef; You, Long

被引用次数: 7

2018

In situ ferromagnetic resonance capability on a polarized neutron reflectometry beamline

JOURNAL OF APPLIED CRYSTALLOGRAPHY

Kostylev, Mikhail; Causer, Grace L.; Lambert, Charles-Henri; Schefer, Thomas; Weiss, Charles; Callori, Sara J.; Salahuddin, Sayeef; Wang, Xiaolin L.; Klose, Frank

被引用次数: 6

2018

Ferroelectric negative capacitance domain dynamics

JOURNAL OF APPLIED PHYSICS

Hoffmann, Michael; Khan, Asif Islam; Serrao, Claudy; Lu, Zhongyuan; Salahuddin, Sayeef; Pesic, Milan; Slesazek, Stefan; Schroeder, Uwe; Mikolajick, Thomas

被引用次数: 75

2018

Differential voltage amplification from ferroelectric negative capacitance (vol 111, 253501, 2017)

APPLIED PHYSICS LETTERS

Khan, Asif I.; Hoffmann, Michael; Chatterjee, Korok; Lu, Zhongyuan; Xu, Ruijuan; Serrao, Claudy; Smith, Samuel; Martin, Lane W.; Hu, Chenming; Ramesh, Ramamoorthy; Salahuddin, Sayeef

被引用次数: 0

2018

Electrically induced, non-volatile, metal insulator transition in a ferroelectric-controlled MoS₂ transistor

APPLIED PHYSICS LETTERS

Lu, Zhongyuan; Serrao, Claudy; Khan, Asif I.; Clarkson, James D.; Wong, Justin C.; Ramesh, Ramamoorthy; Salahuddin, Sayeef

被引用次数: 19

2018

Experimental Evidence of Chiral Ferrimagnetism in Amorphous GdCo Films

ADVANCED MATERIALS

Streubel, Robert; Lambert, Charles-Henri; Kent, Noah; Ercius, Peter; N'Diaye, Alpha T.; Ophus, Colin; Salahuddin, Sayeef; Fischer, Peter

被引用次数: 54

2018

The era of hyper-scaling in electronics

NATURE ELECTRONICS

Salahuddin, Sayeef; Ni, Kai; Datta, Suman

被引用次数: 479

2018

NCFET Design Considering Maximum Interface Electric Field

IEEE ELECTRON DEVICE LETTERS

Agarwal, Harshit; Kushwaha, Pragya; Lin, Yen-Kai; Kao, Ming-Yen; Liao, Yu-Hung; Duarte, Juan-Pablo; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 32

2018

Modeling of Advanced RF Bulk FinFETs

IEEE ELECTRON DEVICE LETTERS

Kushwaha, P.; Agarwal, H.; Lin, Y. -K.; Kao, M. -Y.; Duarte, J. -P.; Chang, H. -L.; Wong, W.; Fan, J.; Xiayu; Chauhan, Y. S.; Salahuddin, S.; Hu, C.

被引用次数: 13

2018

Improved Subthreshold Swing and Short Channel Effect in FDSOI n-Channel Negative Capacitance Field Effect Transistors

IEEE ELECTRON DEVICE LETTERS

Kwon, Daewoong; Chatterjee, Korok; Tan, Ava J.; Yadav, Ajay K.; Zhou, Hong; Sachid, Angada B.; dos Reis, Roberto; Hu, Chenming; Salahuddin, Sayeef

被引用次数: 145

2018

A Nitrided Interfacial Oxide for Interface State Improvement in Hafnium Zirconium Oxide-Based Ferroelectric Transistor Technology

IEEE ELECTRON DEVICE LETTERS

Tan, Ava J.; Yadav, Ajay K.; Chatterjee, Korok; Kwon, Daewoong; Kim, Sangwan; Hu, Chenming; Salahuddin, Sayeef

被引用次数: 34

2018

Negative Differential Resistance and Steep Switching in Chevron Graphene Nanoribbon Field-Effect Transistors

IEEE ELECTRON DEVICE LETTERS

Smith, Samuel; Llinas, Juan-Pablo; Bokor, Jeffrey; Salahuddin, Sayeef

被引用次数: 19

2018

Variation Caused by Spatial Distribution of Dielectric and Ferroelectric Grains in a Negative Capacitance Field-Effect Transistor

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kao, Ming-Yen; Sachid, Angada B.; Lin, Yen-Kai; Liao, Yu-Hung; Agarwal, Harshit; Kushwaha, Pragya; Duarte, Juan Pablo; Chang, Huan-Lin; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 33

2018

Engineering Negative Differential Resistance in NCFETs for Analog Applications

IEEE TRANSACTIONS ON ELECTRON DEVICES

Agarwal, Harshit; Kushwaha, Pragya; Duarte, Juan Pablo; Lin, Yen-Kai; Sachid, Angada B.; Kao, Ming-Yen; Chang, Huan-Lin; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 83

2018

Designing 0.5 V 5-nm HP and 0.23 V 5-nm LP NC-FinFETs With Improved IOFF Sensitivity in Presence of Parasitic Capacitance

IEEE TRANSACTIONS ON ELECTRON DEVICES

Agarwal, Harshit; Kushwaha, Pragya; Duarte, Juan Pablo; Lin, Yen-Kai; Sachid, Angada B.; Chang, Huan-Lin; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 25

2018

New Mobility Model for Accurate Modeling of Transconductance in FDSOI MOSFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lin, Yen-Kai; Kushwaha, Pragya; Duarte, Juan Pablo; Chang, Huan-Lin; Agarwal, Harshit; Khandelwal, Sourabh; Sachid, Angada B.; Harter, Michael; Watts, Josef; Chauhan, Yogesh Singh; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 2

2018

Multidomain Phase-Field Modeling of Negative Capacitance Switching Transients

IEEE TRANSACTIONS ON ELECTRON DEVICES

Smith, Samuel; Chatterjee, Korok; Salahuddin, Sayeef

被引用次数: 23

2018

One-Dimensional Spin Channel in Two-Dimensional Transition Metal Dichalcogenide Heterostructures

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Mishra, Varun; Salahuddin, Sayeef

被引用次数: 5

2018

Voltage-driven, local, and efficient excitation of nitrogen-vacancy centers in diamond

SCIENCE ADVANCES

Labanowski, Dominic; Bhallamudi, Vidya Praveen; Guo, Qiaochu; Purser, Carola M.; McCullian, Brendan A.; Hammel, P. Chris; Salahuddin, Sayeef

被引用次数: 48

2018

Spin wave generation by surface acoustic waves

JOURNAL OF APPLIED PHYSICS

Li, Xu; Labanowski, Dominic; Salahuddin, Sayeef; Lynch, Christopher S.

被引用次数: 37

2017

Differential voltage amplification from ferroelectric negative capacitance

APPLIED PHYSICS LETTERS

Khan, Asif I.; Hoffmann, Michael; Chatterjee, Korok; Lu, Zhongyuan; Xu, Ruijuan; Serrao, Claudy; Smith, Samuel; Martin, Lane W.; Hu, Chenming; Ramesh, Ramamoorthy; Salahuddin, Sayeef

被引用次数: 32

2017

Effect of magnetoelastic film thickness on power absorption in acoustically driven ferromagnetic resonance

APPLIED PHYSICS LETTERS

Labanowski, D.; Jung, A.; Salahuddin, S.

被引用次数: 19

2017

Single shot ultrafast all optical magnetization switching of ferromagnetic Co/Pt multilayers

APPLIED PHYSICS LETTERS

Gorchon, Jon; Lambert, Charles-Henri; Yang, Yang; Pattabi, Akshay; Wilson, Richard B.; Salahuddin, Sayeef; Bokor, Jeffrey	2017
被引用次数: 66	
Stabilization of ferroelectric phase in tungsten capped Hf_{0.8}Zr_{0.2}O₂	
APPLIED PHYSICS LETTERS	
Karbasian, Golnaz; dos Reis, Roberto; Yadav, Ajay K.; Tan, Ava J.; Hu, Chenming; Salahuddin, Sayeef	
被引用次数: 71	2017
Spin-orbit torque switching of ultralarge-thickness ferrimagnetic GdFeCo	
PHYSICAL REVIEW B	
Roschewsky, Niklas; Lambert, Charles-Henri; Salahuddin, Sayeef	
被引用次数: 82	2017
Electric current induced ultrafast demagnetization	
PHYSICAL REVIEW B	
Wilson, Richard B.; Yang, Yang; Gorchon, Jon; Lambert, Charles-Henri; Salahuddin, Sayeef; Bokor, Jeffrey	
被引用次数: 22	2017
Ultrafast magnetic switching of GdFeCo with electronic heat currents	
PHYSICAL REVIEW B	
Wilson, R. B.; Gorchon, Jon; Yang, Yang; Lambert, Charles-Henri; Salahuddin, Sayeef; Bokor, Jeffrey	
被引用次数: 45	2017
High Speed Epitaxial Perovskite Memory on Flexible Substrates	
ADVANCED MATERIALS	
Bakaul, Saidur R.; Serrao, Claudy R.; Lee, Oukjae; Lu, Zhongyuan; Yadav, Ajay; Carraro, Carlo; Maboudian, Roya; Ramesh, Ramamoorthy; Salahuddin, Sayeef	
被引用次数: 81	2017
Sustained Sub-60 mV/decade Switching via the Negative Capacitance Effect in MoS₂ Transistors	
NANO LETTERS	
McGuire, Felicia A.; Lin, Yuh-Chen; Price, Katherine; Rayner, G. Bruce; Khandelwal, Sourabh; Salahuddin, Sayeef; Franklin, Aaron D.	
被引用次数: 270	2017
Intrinsic Limits to Contact Resistivity in Transition Metal Dichalcogenides	
IEEE ELECTRON DEVICE LETTERS	
Mishra, Varun; Salahuddin, Sayeef	
被引用次数: 9	2017
Implementing p-bits With Embedded MTJ	
IEEE ELECTRON DEVICE LETTERS	
Camsari, Kerem Yunus; Salahuddin, Sayeef; Datta, Supriyo	
被引用次数: 133	2017

Self-Aligned, Gate Last, FDSOI, Ferroelectric Gate Memory Device With 5.5-nm Hf_{0.8}Zr_{0.2}O₂, High Endurance and Breakdown Recovery

IEEE ELECTRON DEVICE LETTERS

Chatterjee, Korok; Kim, Sangwan; Karbasian, Golnaz; Tan, Ava J.; Yadav, Ajay K.; Khan, Asif I.; Hu, Chenming; Salahuddin, Sayeef

被引用次数: 83

2017

Intrinsic Speed Limit of Negative Capacitance Transistors

IEEE ELECTRON DEVICE LETTERS

Chatterjee, Korok; Rosner, Alexander John; Salahuddin, Sayeef

被引用次数: 69

2017

Work Function Engineering for Performance Improvement in Leaky Negative Capacitance FETs

IEEE ELECTRON DEVICE LETTERS

Khan, Asif Islam; Radhakrishna, Ujwal; Salahuddin, Sayeef; Antoniadis, Dimitri

被引用次数: 65

2017

Impact of Parasitic Capacitance and Ferroelectric Parameters on Negative Capacitance FinFET Characteristics

IEEE ELECTRON DEVICE LETTERS

Khandelwal, Sourabh; Duarte, Juan Pablo; Khan, Asif Islam; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 77

2017

Modeling of Back-Gate Effects on Gate-Induced Drain Leakage and Gate Currents in UTB SOI MOSFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lin, Yen-Kai; Kushwaha, Pragya; Agarwal, Harshit; Chang, Huan-Lin; Duarte, Juan Pablo; Sachid, Angada B.; Khandelwal, Sourabh; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 9

2017

Compact Modeling Source-to-Drain Tunneling in Sub-10-nm GAA FinFET With Industry Standard Model

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lin, Yen-Kai; Duarte, Juan Pablo; Kushwaha, Pragya; Agarwal, Harshit; Chang, Huan-Lin; Sachid, Angada; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 16

2017

Enabling Energy-Efficient Nonvolatile Computing With Negative Capacitance FET

IEEE TRANSACTIONS ON ELECTRON DEVICES

Li, Xueqing; Sampson, John; Khan, Asif; Ma, Kaisheng; George, Sumitha; Aziz, Ahmedullah; Gupta, Sumeet Kumar; Salahuddin, Sayeef; Chang, Meng-Fan; Datta, Suman; Narayanan, Vijaykrishnan

被引用次数: 68

2017

A Predictive Tunnel FET Compact Model With Atomistic Simulation Validation

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lin, Yen-Kai; Khandelwal, Sourabh; Duarte, Juan Pablo; Chang, Huan-Lin; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 23

2017

Ultrafast magnetization reversal by picosecond electrical pulses

SCIENCE ADVANCES

Yang, Yang; Wilson, Richard B.; Gorchon, Jon; Lambert, Charles-Henri; Salahuddin, Sayeef; Bokor, Jeffrey

被引用次数: 141

2017

Hidden Magnetic States Emergent Under Electric Field, In A Room Temperature Composite Magnetoelectric Multiferroic

SCIENTIFIC REPORTS

Clarkson, J. D.; Fina, I.; Liu, Z. Q.; Lee, Y.; Kim, J.; Frontera, C.; Cordero, K.; Wisotzki, S.; Sanchez, F.; Sort, J.; Hsu, S. L.; Ko, C.; Aballe, L.; Foerster, M.; Wu, J.; Christen, H. M.; Heron, J. T.; Schlom, D. G.; Salahuddin, S.; Kioussis, N.; Fontcuberta, J.; Marti, X.; Ramesh, R.

被引用次数: 27

2017

Spin-orbit torques in ferrimagnetic GdFeCo alloys

APPLIED PHYSICS LETTERS

Roschewsky, Niklas; Matsumura, Tomoya; Cheema, Suraj; Hellman, Frances; Kato, Takeshi; Iwata, Satoshi; Salahuddin, Sayeef

被引用次数: 93

2016

Power absorption in acoustically driven ferromagnetic resonance

APPLIED PHYSICS LETTERS

Labanowski, D.; Jung, A.; Salahuddin, S.

被引用次数: 80

2016

Mechanical back-action of a spin-wave resonance in a magnetoelastic thin film on a surface acoustic wave

PHYSICAL REVIEW B

Gowtham, P. G.; Labanowski, D.; Salahuddin, S.

被引用次数: 27

2016

A 60-nm-thick enhancement mode In_{0.65}Ga_{0.35}As/InAs/In_{0.65}Ga_{0.35}As high-electron-mobility transistor fabricated using Au/Pt/Ti non-annealed ohmic technology for low-power logic applications

APPLIED PHYSICS EXPRESS

Fatah, Faiz Aizad; Lin, Yueh-Chin; Liu, Ren-Xuan; Yang, Kai-Chun; Lin, Tai-We; Hsu, Heng-Tung; Yang, Jung-Hsiang; Miyamoto, Yasuyuki; Iwai, Hiroshi; Hu, Chenming Calvin; Salahuddin, Sayeef; Chang, Edward Yi

被引用次数: 2

2016

Deterministic Spin-Orbit Torque Switching of a Perpendicularly Polarized Magnet Using Wedge Shape of the Magnet

SPIN

Bhowmik, Debanjan; Salahuddin, Sayeef

被引用次数: 1

2016

Direct Observation of Negative Capacitance in Polycrystalline Ferroelectric HfO₂

ADVANCED FUNCTIONAL MATERIALS

Hoffmann, Michael; Pesic, Milan; Chatterjee, Korok; Khan, Asif I.; Salahuddin, Sayeef; Slesazeck, Stefan; Schroeder, Uwe; Mikolajick, Thomas

被引用次数: 249

2016

Surface states in a monolayer MoS₂ transistor

JOURNAL OF MATERIALS RESEARCH

Lu, Zhongyuan; Lee, Oukjae; Wong, Justin C.; Salahuddin, Sayeef

被引用次数: 11

2016

Unified Compact Model Covering Drift-Diffusion to Ballistic Carrier Transport

IEEE ELECTRON DEVICE LETTERS

Khandelwal, Sourabh; Agarwal, Harshit; Kushwaha, Pragya; Duarte, Juan Pablo; Medury, Aditya; Chauhan, Yogesh S.; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 16

2016

Negative Capacitance in Short-Channel FinFETs Externally Connected to an Epitaxial Ferroelectric Capacitor

IEEE ELECTRON DEVICE LETTERS

Khan, Asif Islam; Chatterjee, Korok; Duarte, Juan Pablo; Lu, Zhongyuan; Sachid, Angada; Khandelwal, Sourabh; Ramesh, Ramamoorthy; Hu, Chenming; Salahuddin, Sayeef

被引用次数: 208

2016

Analysis and Compact Modeling of Negative Capacitance Transistor with High ON-Current and Negative Output Differential Resistance-Part I: Model Description

IEEE TRANSACTIONS ON ELECTRON DEVICES

Pahwa, Girish; Dutta, Tapas; Agarwal, Amit; Khandelwal, Sourabh; Salahuddin, Sayeef; Hu, Chenming; Chauhan, Yogesh Singh

被引用次数: 77

2016

Analysis and Compact Modeling of Negative Capacitance Transistor with High ON-Current and Negative Output Differential Resistance-Part II: Model Validation

IEEE TRANSACTIONS ON ELECTRON DEVICES

Pahwa, Girish; Dutta, Tapas; Agarwal, Amit; Khandelwal, Sourabh; Salahuddin, Sayeef; Hu, Chenming; Chauhan, Yogesh Singh

被引用次数: 134

2016

Negative Capacitance Behavior in a Leaky Ferroelectric

IEEE TRANSACTIONS ON ELECTRON DEVICES

Khan, Asif Islam; Radhakrishna, Ujwal; Chatterjee, Korok; Salahuddin, Sayeef; Antoniadis, Dimitri A.

被引用次数: 112

2016

Effects of the Variation of Ferroelectric Properties on Negative Capacitance FET Characteristics

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lin, Cheng-I; Khan, Asif Islam; Salahuddin, Sayeef; Hu, Chenming

Single crystal functional oxides on silicon

NATURE COMMUNICATIONS

Bakaul, Saidur Rahman; Serrao, Claudy Rayan; Lee, Michelle; Yeung, Chun Wing; Sarker, Asis; Hsu, Shang-Lin; Yadav, Ajay Kumar; Dedon, Liv; You, Long; Khan, Asif Islam; Clarkson, James David; Hu, Chenming; Ramesh, Ramamoorthy; Salahuddin, Sayeef

被引用次数: 183

2016

Flexible spin-orbit torque devices

APPLIED PHYSICS LETTERS

Lee, OukJae; You, Long; Jang, Jaewon; Subramanian, Vivek; Salahuddin, Sayeef

被引用次数: 28

2015

Direct optical detection of current induced spin accumulation in metals by magnetization-induced second harmonic generation

APPLIED PHYSICS LETTERS

Pattabi, A.; Gu, Z.; Gorchon, J.; Yang, Y.; Finley, J.; Lee, O. J.; Raziq, H. A.; Salahuddin, S.; Bokor, J.

被引用次数: 54

2015

Highly crystalline MoS2 thin films grown by pulsed laser deposition

APPLIED PHYSICS LETTERS

Serrao, Claudy R.; Diamond, Anthony M.; Hsu, Shang-Lin; You, Long; Gadgil, Sushant; Clarkson, James; Carraro, Carlo; Maboudian, Roya; Hu, Chenming; Salahuddin, Sayeef

被引用次数: 123

2015

Memory leads the way to better computing

NATURE NANOTECHNOLOGY

Wong, H. -S. Philip; Salahuddin, Sayeef

被引用次数: 743

2015

Negative capacitance in a ferroelectric capacitor

NATURE MATERIALS

Khan, Asif Islam; Chatterjee, Korok; Wang, Brian; Drapcho, Steven; You, Long; Serrao, Claudy; Bakaul, Saidur Rahman; Ramesh, Ramamoorthy; Salahuddin, Sayeef

被引用次数: 686

2015

Voltage-Controlled Ferroelastic Switching in Pb(Zr0.2Ti0.8)O3 Thin Films

NANO LETTERS

Khan, Asif Islam; Marti, Xavier; Serrao, Claudy; Ramesh, Ramamoorthy; Salahuddin, Sayeef

被引用次数: 52

2015

Modeling SiGe FinFETs With Thin Fin and Current-Dependent Source/Drain Resistance

IEEE ELECTRON DEVICE LETTERS

Khandelwal, Sourabh; Duarte, Juan Pablo; Medury, Aditya; Chauhan, Yogesh S.; Salahuddin, Sayeef; Hu, Chenming

被引用次数: 8

2015

Screening in Ultrashort (5 nm) Channel MoS₂ Transistors: A Full-Band Quantum Transport Study

IEEE TRANSACTIONS ON ELECTRON DEVICES

Mishra, Varun; Smith, Samuel; Liu, Lei; Zahid, Ferdows; Zhu, Yu; Guo, Hong; Salahuddin, Sayeef

被引用次数: 24

2015

Switching of perpendicularly polarized nanomagnets with spin orbit torque without an external magnetic field by engineering a tilted anisotropy

PROCEEDINGS OF THE NATIONAL ACADEMY OF SCIENCES OF THE UNITED STATES OF AMERICA

You, Long; Lee, OukJae; Bhowmik, Debanjan; Labanowski, Dominic; Hong, Jeongmin; Bokor, Jeffrey; Salahuddin, Sayeef

被引用次数: 282

2015

Large resistivity modulation in mixed-phase metallic systems

NATURE COMMUNICATIONS

Lee, Yeonbae; Liu, Z. Q.; Heron, J. T.; Clarkson, J. D.; Hong, J.; Ko, C.; Biegalski, M. D.; Aschauer, U.; Hsu, S. L.; Nowakowski, M. E.; Wu, J.; Christen, H. M.; Salahuddin, S.; Bokor, J. B.; Spaldin, N. A.; Schlom, D. G.; Ramesh, R.

被引用次数: 174

2015

Deterministic Domain Wall Motion Orthogonal To Current Flow Due To Spin Orbit Torque

SCIENTIFIC REPORTS

Bhowmik, Debanjan; Nowakowski, Mark E.; You, Long; Lee, OukJae; Keating, David; Wong, Mark; Bokor, Jeffrey; Salahuddin, Sayeef

被引用次数: 70

2015

Magnetic domain-wall motion twisted by nanoscale probe-induced spin transfer

PHYSICAL REVIEW B

Wang, J.; Xie, L. S.; Wang, C. S.; Zhang, H. Z.; Shu, L.; Bai, J.; Chai, Y. S.; Zhao, X.; Nie, J. C.; Cao, C. B.; Gu, C. Z.; Xiong, C. M.; Sun, Y.; Shi, J.; Salahuddin, S.; Xia, K.; Nan, C. W.; Zhang, J. X.

被引用次数: 17

2014

Deterministic switching of ferromagnetism at room temperature using an electric field

NATURE

Heron, J. T.; Bosse, J. L.; He, Q.; Gao, Y.; Trassin, M.; Ye, L.; Clarkson, J. D.; Wang, C.; Liu, Jian; Salahuddin, S.; Ralph, D. C.; Schlom, D. G.; Iniguez, J.; Huey, B. D.; Ramesh, R.

被引用次数: 613

2014

The effects of strain relaxation on the dielectric properties of epitaxial ferroelectric Pb(Zr_{0.2}Ti_{0.8})TiO₃ thin films

APPLIED PHYSICS LETTERS

Khan, Asif Islam; Yu, Pu; Trassin, Morgan; Lee, Michelle J.; You, Long; Salahuddin, Sayeef

被引用次数: 12

2014

Spin Hall effect clocking of nanomagnetic logic without a magnetic field

NATURE NANOTECHNOLOGY

Bhowmik, Debanjan; You, Long; Salahuddin, Sayeef

Room-temperature antiferromagnetic memory resistor

NATURE MATERIALS

Marti, X.; Fina, I.; Frontera, C.; Liu, Jian; Wadley, P.; He, Q.; Paull, R. J.; Clarkson, J. D.; Kudrnovsky, J.; Turek, I.; Kunes, J.; Yi, D.; Chu, J-H.; Nelson, C. T.; You, L.; Arenholz, E.; Salahuddin, S.; Fontcuberta, J.; Jungwirth, T.; Ramesh, R.

被引用次数: 580

2014

Room-Temperature Negative Capacitance in a Ferroelectric Dielectric Super lattice Heterostructure

NANO LETTERS

Gao, Weiwei; Khan, Asif; Marti, Xavi; Nelson, Chris; Serrao, Claudy; Ravichandran, Jayakanth; Ramesh, Ramamoorthy; Salahuddin, Sayeef

被引用次数: 134

2014

Real-Time Observation of Local Strain Effects on Nonvolatile Ferroelectric Memory Storage Mechanisms

NANO LETTERS

Winkler, Christopher R.; Jablonski, Michael L.; Ashraf, Khalid; Damodaran, Anoop R.; Jambunathan, Karthik; Hart, James L.; Wen, Jianguo G.; Miller, Dean J.; Martin, Lane W.; Salahuddin, Sayeef; Taheri, Mitra L.

被引用次数: 18

2014

Gate Recessed Quasi-Normally OFF Al₂O₃/AlGa_N/Ga_N MIS-HEMT With Low Threshold Voltage Hysteresis Using PEALD Al_N Interfacial Passivation Layer

IEEE ELECTRON DEVICE LETTERS

Hsieh, Ting-En; Chang, Edward Yi; Song, Yi-Zuo; Lin, Yueh-Chin; Wang, Huan-Chung; Liu, Shin-Chien; Salahuddin, Sayeef; Hu, Chenming Calvin

被引用次数: 120

2014

Electrical Characteristics of n, p-In_{0.53}Ga_{0.47}As MOSCAPs With In Situ PEALD-Al_N Interfacial Passivation Layer

IEEE TRANSACTIONS ON ELECTRON DEVICES

Quang Ho Luc; Chang, Edward Yi; Hai Dang Trinh; Lin, Yueh Chin; Hong Quan Nguyen; Wong, Yuen Yee; Huy Binh Do; Salahuddin, Sayeef; Hu, Chenming Calvin

被引用次数: 39

2014

Nature of magnetic domains in an exchange coupled BiFeO₃/CoFe heterostructure

APPLIED PHYSICS LETTERS

Qiu, D. Y.; Ashraf, K.; Salahuddin, S.

被引用次数: 27

2013

Progress, Challenges, and Opportunities in Two-Dimensional Materials Beyond Graphene

ACS NANO

Butler, Sheneve Z.; Hollen, Shawna M.; Cao, Linyou; Cui, Yi; Gupta, Jay A.; Gutierrez, Humberto R.; Heinz, Tony F.; Hong, Seung Sae; Huang, Jiaying; Ismach, Ariel F.; Johnston-Halperin, Ezekiel; Kuno, Masaru;

Plashnitsa, Vladimir V.; Robinson, Richard D.; Ruoff, Rodney S.; Salahuddin, Sayeef; Shan, Jie; Shi, Li; Spencer, Michael G.; Terrones, Mauricio; Windl, Wolfgang; Goldberger, Joshua E.

被引用次数: 4509

2013

Electrothermal Analysis of Spin-Transfer-Torque Random Access Memory Arrays

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Chatterjee, Subho; Salahuddin, Sayeef; Kumar, Satish; Mukhopadhyay, Saibal

被引用次数: 0

2013

Phenomenological Compact Model for QM Charge Centroid in Multigate FETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Venugopalan, Sriramkumar; Karim, Muhammed A.; Salahuddin, Sayeef; Niknejad, Ali M.; Hu, Chenming Calvin

被引用次数: 17

2013

Ballistic I-V Characteristics of Short-Channel Graphene Field-Effect Transistors: Analysis and Optimization for Analog and RF Applications

IEEE TRANSACTIONS ON ELECTRON DEVICES

Ganapathi, Kartik; Yoon, Youngki; Lundstrom, Mark; Salahuddin, Sayeef

被引用次数: 21

2013

High Performance Molybdenum Disulfide Amorphous Silicon Heterojunction Photodetector

SCIENTIFIC REPORTS

Esmaeili-Rad, Mohammad R.; Salahuddin, Sayeef

被引用次数: 138

2013

Phase field model of domain dynamics in micron scale, ultrathin ferroelectric films: Application for multiferroic bismuth ferrite

JOURNAL OF APPLIED PHYSICS

Ashraf, Khalid; Salahuddin, Sayeef

被引用次数: 10

2012

Zener tunneling: Congruence between semi-classical and quantum ballistic formalisms

JOURNAL OF APPLIED PHYSICS

Ganapathi, Kartik; Salahuddin, Sayeef

被引用次数: 6

2012

Effect of anti-ferromagnet surface moment density on the hysteresis properties of exchange coupled antiferromagnet-ferromagnet systems: The case of bismuth-ferrite

JOURNAL OF APPLIED PHYSICS

Ashraf, Khalid; Salahuddin, Sayeef

被引用次数: 4

2012

Dissipative transport in rough edge graphene nanoribbon tunnel transistors

APPLIED PHYSICS LETTERS

Yoon, Youngki; Salahuddin, Sayeef

被引用次数: 29

2012

Non-volatile spin switch for Boolean and non-Boolean logic

APPLIED PHYSICS LETTERS

Datta, Supriyo; Salahuddin, Sayeef; Behin-Aein, Behtash

被引用次数: 116

2012

Experimental evidence of ferroelectric negative capacitance in nanoscale heterostructures (vol 99, 113501, 2011)

APPLIED PHYSICS LETTERS

Khan, Asif Islam; Bhowmik, Debanjan; Yu, Pu; Kim, Sung Joo; Pan, Xiaoqing; Ramesh, Ramamoorthy; Salahuddin, Sayeef

被引用次数: 2

2012

Dense Electron System from Gate-Controlled Surface Metal-Insulator Transition

NANO LETTERS

Liu, Kai; Fu, Deyi; Cao, Jinbo; Suh, Joonki; Wang, Kevin X.; Cheng, Chun; Ogletree, D. Frank; Guo, Hua; Sengupta, Shamashis; Khan, Asif; Yeung, Chun Wing; Salahuddin, Sayeef; Deshmukh, Mandar M.; Wu, Junqiao

被引用次数: 64

2012

Impact of Self-Heating on Reliability of a Spin-Torque-Transfer RAM Cell

IEEE TRANSACTIONS ON ELECTRON DEVICES

Chatterjee, Subho; Salahuddin, Sayeef; Kumar, Satish; Mukhopadhyay, Saibal

被引用次数: 9

2012

Voltage Asymmetry of Spin-Transfer Torques

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Datta, Deepanjan; Behin-Aein, Behtash; Datta, Supriyo; Salahuddin, Sayeef

被引用次数: 106

2012

Built-in and induced polarization across LaAlO₃/SrTiO₃ heterojunctions

NATURE PHYSICS

Singh-Bhalla, Guneeta; Bell, Christopher; Ravichandran, Jayakanth; Siemons, Wolter; Hikita, Yasuyuki; Salahuddin, Sayeef; Hebard, Arthur F.; Hwang, Harold Y.; Ramesh, Ramamoorthy

被引用次数: 184

2011

Electric-Field-Induced Magnetization Reversal in a Ferromagnet-Multiferroic Heterostructure

PHYSICAL REVIEW LETTERS

Heron, J. T.; Trassin, M.; Ashraf, K.; Gajek, M.; He, Q.; Yang, S. Y.; Nikonov, D. E.; Chu, Y-H.; Salahuddin, S.; Ramesh, R.

被引用次数: 429

2011

Experimental evidence of ferroelectric negative capacitance in nanoscale heterostructures

APPLIED PHYSICS LETTERS

Khan, Asif Islam; Bhowmik, Debanjan; Yu, Pu; Kim, Sung Joo; Pan, Xiaoqing; Ramesh, Ramamoorthy; Salahuddin, Sayeef

被引用次数: 307

2011

Role of phonon scattering in graphene nanoribbon transistors: Nonequilibrium Green's function method with real space approach

APPLIED PHYSICS LETTERS

Yoon, Youngki; Nikonov, Dmitri E.; Salahuddin, Sayeef

被引用次数: 34

2011

How Good Can Monolayer MoS₂ Transistors Be?

NANO LETTERS

Yoon, Youngki; Ganapathi, Kartik; Salahuddin, Sayeef

被引用次数: 1482

2011

Heterojunction Vertical Band-to-Band Tunneling Transistors for Steep Subthreshold Swing and High ON Current

IEEE ELECTRON DEVICE LETTERS

Ganapathi, Kartik; Salahuddin, Sayeef

被引用次数: 77

2011

Performance analysis of carbon-based tunnel field-effect transistors for high frequency and ultralow power applications

APPLIED PHYSICS LETTERS

Yoon, Youngki; Kim, Sung Hwan; Salahuddin, Sayeef

被引用次数: 4

2010

Analysis of InAs vertical and lateral band-to-band tunneling transistors: Leveraging vertical tunneling for improved performance

APPLIED PHYSICS LETTERS

Ganapathi, Kartik; Yoon, Youngki; Salahuddin, Sayeef

被引用次数: 84

2010

Barrier-free tunneling in a carbon heterojunction transistor

APPLIED PHYSICS LETTERS

Yoon, Youngki; Salahuddin, Sayeef

被引用次数: 32

2010

Inverse temperature dependence of subthreshold slope in graphene nanoribbon tunneling transistors

APPLIED PHYSICS LETTERS

Yoon, Youngki; Salahuddin, Sayeef

被引用次数: 13

2010

Proposal for an all-spin logic device with built-in memory

NATURE NANOTECHNOLOGY

Behin-Aein, Behtash; Datta, Deepanjan; Salahuddin, Sayeef; Datta, Supriyo

Dual-Source-Line-Bias Scheme to Improve the Read Margin and Sensing Accuracy of STT RAM in Sub-90-nm Nodes

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Chatterjee, Subho; Salahuddin, Sayeef; Mukhopadhyay, Saibal

被引用次数: 2

2010

Design Paradigm for Robust Spin-Torque Transfer Magnetic RAM (STT MRAM) From Circuit/Architecture Perspective

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Li, Jing; Ndai, Patrick; Goel, Ashish; Salahuddin, Sayeef; Roy, Kaushik

被引用次数: 125

2010

Ultrathin compound semiconductor on insulator layers for high-performance nanoscale transistors

NATURE

Ko, Hyunhyub; Takei, Kuniharu; Kapadia, Rehan; Chuang, Steven; Fang, Hui; Leu, Paul W.; Ganapathi, Kartik; Plis, Elena; Kim, Ha Sul; Chen, Szu-Ying; Madsen, Morten; Ford, Alexandra C.; Chueh, Yu-Lun; Krishna, Sanjay; Salahuddin, Sayeef; Javey, Ali

被引用次数: 418

2010

Switching Energy of Ferromagnetic Logic Bits

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Behin-Aein, Behtash; Salahuddin, Sayeef; Datta, Supriyo

被引用次数: 73

2009

Use of negative capacitance to provide voltage amplification for low power nanoscale devices

NANO LETTERS

Salahuddin, Sayeef; Dattat, Supriyo

被引用次数: 1854

2008

Interacting systems for self-correcting low power switching

APPLIED PHYSICS LETTERS

Salahuddin, Sayeef; Datta, Supriyo

被引用次数: 96

2007

Self-consistent simulation of quantum transport and magnetization dynamics in spin-torque based devices

APPLIED PHYSICS LETTERS

Salahuddin, Sayeef; Datta, Supriyo

被引用次数: 30

2006

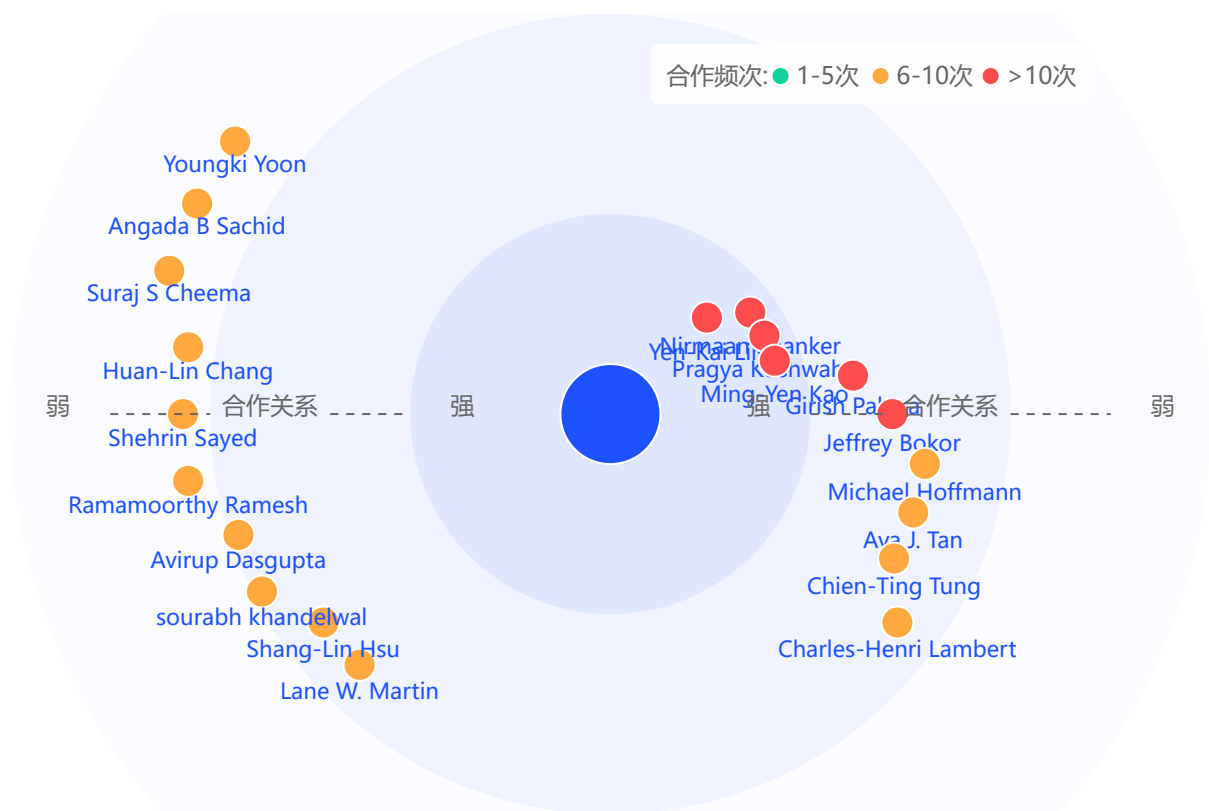
High-frequency performance projections for ballistic carbon-nanotube transistors

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Hasan, S; Salahuddin, S; Vaidyanathan, M; Alam, AA

被引用次数: 64	2006
Transport effects on signal propagation in quantum wires	
IEEE TRANSACTIONS ON ELECTRON DEVICES	
Salahuddin, S; Lundstrom, M; Datta, S	
被引用次数: 154	2005
Reducing signal-bias from MAD estimated noise level for DCT speech enhancement	
SIGNAL PROCESSING	
Hasan, MK; Salahuddin, S; Khan, MR	
被引用次数: 6	2004
A modified A priori SNR for speech enhancement using spectral subtraction rules	
IEEE SIGNAL PROCESSING LETTERS	
Hasan, MK; Salahuddin, S; Khan, MR	
被引用次数: 89	2004
Soft thresholding for DCT speech enhancement	
ELECTRONICS LETTERS	
Salahuddin, S; Al Islam, SZ; Hasan, MK; Khan, MR	
被引用次数: 21	2002
A Density Metric for Semiconductor Technology [Point of View]	
Proc.IEEE	
H.-S. Philip Wong; Kerem Akarvardar; Dimitri Antoniadis; Jeffrey Bokor; Chenming Hu; Tsu-Jae King-Liu; Subhasish Mitra; James D. Plummer; Sayeef Salahuddin	
被引用次数: 0	
Negative Capacitance Transistors	
Proc.IEEE	
Justin C. Wong; Sayeef Salahuddin	
被引用次数: 0	

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